

Full Development History: Rust Task Manager

Comprehensive transcript of development, troubleshooting, and UI/UX iteration.

USER PROMPT

Initial request to build a Linux Task Manager with X11 GUI, Neofetch, and resource monitoring, transitioning from Llama 3 output.

ASSISTANT RESPONSE

Identified hallucinations in Llama output and introduced the GTK4 + sysinfo stack. Provided the first working skeleton using Notebook tabs.

USER PROMPT

Reported error[E0382]: borrow of moved value: `entry`

ASSISTANT RESPONSE

Explained the Borrow Checker and ownership move into closures. Introduced `.clone()` for GTK widgets.

USER PROMPT

Reported Graphics Errors: libEGL warning: failed to get driver name for fd -1

ASSISTANT RESPONSE

Identified VM/WSL2 graphics acceleration issues. Provided `LIBGL_ALWAYS_SOFTWARE=1` bypass.

USER PROMPT

Requested to make the "Add Task" button functional and subsequently add a "Delete" button for each row.

ASSISTANT RESPONSE

Implemented nested closures and signal connectors. Solved the "non-child" removal warning by using `.unparent()` and `.parent()` logic.

USER PROMPT

Reported error: this file contains an unclosed delimiter

ASSISTANT RESPONSE

Debugged bracket nesting in closures. Provided a consolidated `main.rs` to ensure syntax integrity.

USER PROMPT

Requested a Windows-style Task Manager with a process table (including Virtual Memory) and a live performance graph with X/Y axes.

ASSISTANT RESPONSE

Implemented the `ColumnView` with a custom `GObject ProcessWrapper`. Added Cairo drawing logic for the graph with padding and coordinate offsets for labels. Fixed the `margin_all` builder error.

FINAL REFERENCE CODE SNIPPET

```
// Sample of the final logic implemented:
graph_area.set_draw_func(move |_, cr, w, h| {
    let padding = 45.0;
    let graph_h = h as f64 - 40.0;
    cr.set_source_rgb(0.01, 0.01, 0.01);
    cr.paint().unwrap();
    // Logic for Y-axis labels and CPU data plotting...
});
```